Analysis 10: Fixed Crossing Delay and Concentration for Weighted Rolling Validation

Aileen Tao

1 Motivation: from exact alignment to fixed crossing delay

Analysis 09 showed that exact crossing alignment concentrates positive rolling-validation (RV) weights on an increasingly small fraction of recent validation times, obstructing Chebyshev concentration on the moving-crossing scale. Analysis 10 studies the tradeoff between fixed crossing delay and concentration:

$$\text{If we allow } \frac{n_{\mathrm{RV},w}}{n_{\mathrm{pop}}} \to \lambda > 1,$$

how much do averaging and concentration improve,

and is that enough for the Chebyshev ratio to vanish?

Analysis 06 established power-weight concentration against an oracle. We reuse its full variance calculation, distinguishing that benchmark from Analysis 09's moving-crossing model.

Let $Y_i = \theta + \xi_i$, with independent $\xi_i \sim N(0,1)$ and $0 < |\theta| < 1$. Model 0 predicts with the training sample mean; Model 1 predicts zero. Both use squared loss. At validation time $i = 2, \ldots, n+1$, the training size is $i-1$. Throughout, the weights $w_i > 0$ form a deterministic sequence independent of the data and θ. The parameter λ is fixed and finite unless a subsequent parameter limit is explicitly stated.

2 General constant crossing ratio

Analysis 09 gives

$$\Delta_{\mathrm{RV},w}(n) = \sum_{i=2}^{n+1} w_i \left(\frac{1}{i-1} - \theta^2 \right), \qquad r_n = \frac{\sum_{i=2}^{n+1} w_i/(i-1)}{\sum_{i=2}^{n+1} w_i}.$$

Dividing by positive total weight preserves the crossing:

$$\frac{\Delta_{\mathrm{RV},w}(n)}{\sum_{i=2}^{n+1} w_i} = r_n - \theta^2, \qquad \Delta_R(n) = \frac{1}{n} - \theta^2.$$

Thus r_n plays the same role for weighted RV that $1/n$ plays for population risk. Write $n_{\mathrm{RV},w}$ and n_{pop} for the first strict negative indices of these differences. Since $r_n \geq 1/n$, the normalized expected weighted-RV difference is always at least as large as the population-risk difference. Thus weighted RV cannot cross earlier than population risk:

$$n_{\mathrm{RV},w} \geq n_{\mathrm{pop}}$$

whenever the weighted crossing exists. Therefore any finite limiting crossing ratio must satisfy

$$\lambda \geq 1.$$

Theorem 2.1 (Constant crossing delay)**.** For fixed $\lambda \geq 1$,

$$nr_n \to \lambda \quad \Longleftrightarrow \quad r_n \sim \frac{\lambda}{n}.$$

Under this condition, $n_{\mathrm{RV},w}$ exists for every $0 < |\theta| < 1$, and

$$n_{\mathrm{pop}} = \lfloor \theta^{-2} \rfloor + 1 \sim \theta^{-2}, \qquad \frac{n_{\mathrm{RV},w}}{n_{\mathrm{pop}}} \to \lambda \qquad (|\theta| \to 0).$$

Interpretation. The value $\lambda = 1$ is the earliest possible asymptotic crossing; $\lambda > 1$ imposes a fixed delay. **Proof.** See Appendix A.2, extending Analysis 09, Appendix B.4.

3 Power weights and the λ-tradeoff

Analysis 06 studied the same power-weight family $w_i = i^\alpha$ under an oracle comparison. Here we use this family to quantify the crossing-delay and averaging tradeoff in the present model.

Proposition 3.1 (Power weights and crossing delay)**.** For $w_i = i^\alpha$ with fixed $\alpha > 0$,

$$\sum_{i=2}^{n+1} w_i \sim \frac{n^{\alpha+1}}{\alpha+1}, \qquad \sum_{i=2}^{n+1} \frac{w_i}{i-1} \sim \frac{n^\alpha}{\alpha}, \qquad \sum_{i=2}^{n+1} w_i^2 \sim \frac{n^{2\alpha+1}}{2\alpha+1}.$$

Consequently,

$$r_n \sim \frac{\alpha+1}{\alpha n} \quad \Longrightarrow \quad nr_n \to \frac{\alpha+1}{\alpha}.$$

Hence, by Theorem 2.1,

$$\lambda = \frac{\alpha+1}{\alpha} = 1 + \frac{1}{\alpha}, \qquad \frac{n_{\mathrm{RV},w}}{n_{\mathrm{pop}}} \to \lambda.$$

Interpretation. The relation

$$\lambda = 1 + \frac{1}{\alpha}$$

shows how the power exponent controls the crossing delay. Equivalently,

$$\alpha = \frac{1}{\lambda - 1},$$

so every fixed delay $\lambda > 1$ can be achieved by an appropriate choice of α. For power weights,

$$n \frac{\sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2} \to \frac{(\alpha+1)^2}{2\alpha+1}.$$

Substituting $\alpha = 1/(\lambda - 1)$ gives

$$\frac{(\alpha+1)^2}{2\alpha+1} = \frac{\lambda^2}{\lambda^2 - 1}.$$

This normalized square-weight factor measures how concentrated the weights are. Values closer to 1 correspond to more evenly spread weights and stronger averaging, while larger values correspond to more concentrated weights and weaker averaging.

The following parameter limits are taken after $n \to \infty$ at fixed $\lambda > 1$. As $\lambda \downarrow 1$,

$$\lambda \downarrow 1 \quad \Longrightarrow \quad \alpha \to \infty, \qquad \lambda^2 - 1 \to 0^+, \qquad \frac{\lambda^2}{\lambda^2 - 1} \to \infty.$$

As λ approaches 1, α becomes large and the power weights put more mass on the latest validation times. The weights are therefore more concentrated, so averaging becomes weaker. This matches Analysis 09: exact alignment ($\lambda = 1$) forces the normalized square-weight factor to diverge.

Conversely,

$$\lambda \to \infty \quad \Longrightarrow \quad \alpha \downarrow 0, \qquad \frac{\lambda^2}{\lambda^2 - 1} = \frac{1}{1 - \lambda^{-2}} \longrightarrow 1.$$

As $\lambda \to \infty$, $\alpha \to 0$, so the power weights become close to equal weights. The weights are more evenly spread, the normalized square-weight factor approaches 1, and averaging becomes stronger. The price is a much later crossing. **Proof.** See Appendix B.1 and Theorem 2.1.

4 Concentration on the moving-crossing scale for fixed $\lambda > 1$

Analysis 09 gives the realized-score lower bound

$$\Xi_{n,w} = \sum_{i=2}^{n+1} w_i \{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\}, \qquad \mathbb{E}\Xi_{n,w} = \left(\sum_{i=2}^{n+1} w_i\right)(r_n - \theta^2),$$

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \geq \frac{4\theta^2 \sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2 (r_n - \theta^2)^2}, \tag{1}$$

when the mean is nonzero (Analysis 09, Appendices D.1–D.2).

Under

$$n r_n \to \lambda \in [1, \infty), \qquad n\theta^2 \to c \in (0, \infty), \qquad |\theta| \to 0,$$

we have

$$r_n \sim \frac{\lambda}{n}, \qquad \theta^2 \sim \frac{c}{n}, \qquad r_n - \theta^2 \sim \frac{\lambda - c}{n} \quad (c \neq \lambda).$$

Substitution into (1) gives

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \geq (1 + o(1)) \frac{4c}{(\lambda - c)^2} \, n \frac{\sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2}.$$

For positive weights, Cauchy–Schwarz gives

$$\left(\sum_{i=2}^{n+1} w_i\right)^2 \leq n \sum_{i=2}^{n+1} w_i^2,$$

so

$$n \frac{\sum_{i=2}^{n+1} w_i^2}{\left(\sum_{i=2}^{n+1} w_i\right)^2} \geq 1.$$

This removes dependence on the specific weights and gives the general lower bound below.

Proposition 4.1 (Positive-weight lower bound)**.** Under the preceding limits and positive deterministic weights,

$$\liminf \frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \geq \frac{4c}{(\lambda - c)^2} > 0, \qquad c \neq \lambda.$$

At $c = \lambda$, the leading normalized mean vanishes; the ratio diverges when the mean is nonzero and is undefined at exact mean-zero points.

Interpretation. For any finite crossing delay λ, the Chebyshev ratio does not vanish on this scale. **Proof.** See Appendix C.2.

For power weights $w_i = i^\alpha$, $\alpha > 0$, the current-model mean is

$$\mathbb{E}\Xi_{n,w} = \left(\sum_{i=2}^{n+1} w_i\right)(r_n - \theta^2) \sim \frac{n^{\alpha+1}}{\alpha+1}\frac{\lambda - c}{n}.$$

Since $\lambda = (\alpha+1)/\alpha$,

$$\mathbb{E}\Xi_{n,w} \sim n^\alpha \left(\frac{1}{\alpha} - \frac{c}{\alpha+1}\right), \qquad c \neq \lambda.$$

The quadratic-noise component is unchanged from Analysis 06, so its variance asymptotic can be reused directly:

$$\mathrm{Var}(\Xi_n^{(\alpha)}) \sim \frac{2n^{2\alpha}}{\alpha+1}.$$

The present model adds the linear-noise term

$$-2\theta \sum_{i=2}^{n+1} i^\alpha \xi_i,$$

whose variance is

$$4\theta^2 \sum_{i=2}^{n+1} i^{2\alpha} \sim \frac{4c}{2\alpha+1} n^{2\alpha}.$$

Therefore,

$$\mathrm{Var}(\Xi_{n,w}) \sim n^{2\alpha}\left(\frac{2}{\alpha+1} + \frac{4c}{2\alpha+1}\right).$$

The first term is the Analysis 06 quadratic component; the second is the additional linear-noise contribution (Appendix C.1).

Combining the mean and variance gives the following result.

Theorem 4.2 (Full moving-crossing ratio for power weights)**.** For fixed $\alpha > 0$, $\lambda = 1 + 1/\alpha$, and the preceding moving-crossing limits with $c \neq \lambda$,

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \to \frac{2\lambda/(\lambda-1) + 4c\lambda^2/(\lambda^2-1)}{(\lambda - c)^2} \in (0, \infty). \tag{2}$$

At $c = \lambda$, the ratio diverges when defined.

Interpretation. For fixed finite $\lambda > 1$, the limit is positive for $c \neq \lambda$, so the Chebyshev ratio does not vanish; at $c = \lambda$, it diverges when defined. **Proof.** See Appendices C.1–C.3.

5 Conclusion and implications

Allowing a fixed crossing delay improves averaging compared with the exact-alignment case studied in Analysis 09, because the weights do not need to be as concentrated on the latest validation times.

However, any fixed finite delay is still not enough to make the variance-to-mean-squared ratio vanish on the moving-crossing scale. The ratio remains positive away from the weighted crossing and becomes unbounded at the crossing itself, when defined.

The main tradeoff is clear: better crossing alignment requires more concentrated weights and weaker averaging, while stronger averaging requires accepting a larger crossing delay.

Appendix

A General crossing theorem

A.1 Population risk, positivity, and monotonicity

An independent new response has variance 1. The sample-mean predictor has estimation variance $1/n$ and no bias, while the zero predictor has squared bias θ^2. Thus their risks are $1 + 1/n$ and $1 + \theta^2$, respectively. Their difference is negative exactly when $n > \theta^{-2}$, giving $n_{\text{pop}} = \lfloor \theta^{-2} \rfloor + 1$. This is the calculation in Analysis 09, Appendix A.1, and applies to either sign of θ.

The one-time expected validation difference is $1/(i-1) - \theta^2$. Linearity of expectation gives the weighted mean in Section 2 without requiring independence of validation losses. Every reciprocal in r_n is at least $1/n$, so $r_n \geq 1/n$ and the weighted strict crossing cannot precede the population crossing. For $n > 1$, the inequality is strict because the earlier terms have positive weights.

Adding the next positive weight gives

$$r_{n+1} = \frac{(\sum_{i=2}^{n+1} w_i) r_n + w_{n+2}/(n+1)}{\sum_{i=2}^{n+1} w_i + w_{n+2}}.$$

Since $r_n \geq 1/n > 1/(n+1)$, this weighted average is strictly smaller than r_n. Thus once $r_n < \theta^2$ occurs, it remains so. Under $nr_n \to \lambda < \infty$, $r_n \to 0$; hence a first strict crossing exists for each fixed $\theta \neq 0$. Also $r_1 = 1 > \theta^2$.

A.2 Epsilon bracketing of the first strict crossing

Because $\lambda > 0$, the identity $r_n/(\lambda/n) = nr_n/\lambda$ proves the equivalence in Theorem 2.1. To prove its crossing conclusion, put $h = \theta^2 \downarrow 0$ and fix $\varepsilon \in (0, 1)$. Set

$$n_- = \left\lfloor \frac{(1-\varepsilon)\lambda}{h} \right\rfloor, \qquad n_+ = \left\lceil \frac{(1+\varepsilon)\lambda}{h} \right\rceil.$$

Both indices tend to infinity and rounding changes either by at most one. Since $nr_n \to \lambda$,

$$\frac{r_{n_-}}{h} \to \frac{1}{1-\varepsilon} > 1, \qquad \frac{r_{n_+}}{h} \to \frac{1}{1+\varepsilon} < 1.$$

For small h, the mean is positive at n_- and strictly negative at n_+. Monotonicity from Appendix A.1 therefore gives $n_- < n_{\text{RV},w} \leq n_+$. Dividing by h^{-1} bounds the lower and upper limits of $hn_{\text{RV},w}$ by $(1-\varepsilon)\lambda$ and $(1+\varepsilon)\lambda$. Letting $\varepsilon \downarrow 0$ yields $hn_{\text{RV},w} \to \lambda$. Since $hn_{\text{pop}} \to 1$, the desired ratio follows. All dependence on θ is through h, so the result holds as $|\theta| \to 0$ from either sign.

B Power sums and reparameterization

B.1 Integral comparisons and indexing shifts

For fixed $\beta > -1$, integral comparison gives

$$\sum_{i=2}^{n+1} i^{\beta} \sim \frac{n^{\beta+1}}{\beta+1}.$$

For $\beta \geq 0$, monotonicity sandwiches the sum between integrals with endpoints shifted by at most one; the endpoint error is $O(n^{\beta}+1) = o(n^{\beta+1})$. For $-1 < \beta < 0$, the decreasing summand has a bounded sum–integral discrepancy, also negligible relative to the divergent integral. Replacing $n+1$ by n in the leading power is harmless since $(n+1)/n \to 1$.

Apply this result with $\beta = \alpha$ and 2α to obtain the total-weight and square-weight sums in Proposition 3.1. For the reciprocal-weight sum,

$$\frac{i^{\alpha}/(i-1)}{i^{\alpha-1}} = \frac{i}{i-1} \to 1.$$

The comparison sum $\sum_{i=2}^{n+1} i^{\alpha-1}$ diverges for $\alpha > 0$. For any fixed tolerance, the relative error in the tail is eventually within that tolerance, and the finite initial part is negligible. Thus the reciprocal-weight sum is asymptotic to n^{α}/α. Dividing by total weight gives $nr_n \to (\alpha+1)/\alpha$; Theorem 2.1 then supplies the crossing ratio.

B.2 Delay and averaging constants

Squaring the total-weight asymptotic and dividing the square-weight sum by it gives

$$n\frac{\sum_{i=2}^{n+1} i^{2\alpha}}{(\sum_{i=2}^{n+1} i^{\alpha})^2} \to \frac{(\alpha+1)^2}{2\alpha+1}.$$

With $\alpha = 1/(\lambda-1)$,

$$\alpha + 1 = \frac{\lambda}{\lambda-1}, \qquad 2\alpha+1 = \frac{\lambda+1}{\lambda-1},$$

so this constant is $\lambda^2/(\lambda^2-1)$. The oracle result from Analysis 06 transforms as

$$\frac{2\alpha^2}{1+\alpha} = \frac{2}{\lambda(\lambda-1)}.$$

Both expressions decrease strictly on $\lambda > 1$. Their derivatives are $-2\lambda/(\lambda^2-1)^2$ and $-2(2\lambda - 1)/\{\lambda^2(\lambda-1)^2\}$, respectively. Their endpoint limits are the ones stated in Section 3. These are limits of constants obtained for each fixed $\alpha > 0$, not a claim that the asymptotics hold uniformly as $\alpha \downarrow 0$. At $\alpha = 0$, Analysis 06 has a vanishing oracle ratio, while Analysis 09's ordinary-RV crossing has a diverging logarithmic delay rather than a finite λ.

C Concentration on the moving-crossing scale

C.1 Transfer of the Analysis 06 variance

Write $S_{i-1} = \sum_{t=1}^{i-1} \xi_t$ in this appendix. Expanding one loss difference gives

$$(Y_i - \bar{Y}_{i-1})^2 - Y_i^2 = \frac{S_{i-1}^2}{(i-1)^2} - 2\xi_i\frac{S_{i-1}}{i-1} - 2\theta\xi_i - \theta^2.$$

The first two terms are exactly the oracle loss difference used in Analysis 06. Its score with validation endpoint $n+1$ is therefore related to the present power-weight score by

$$\Xi_{n,w} = \Xi_{n+1}^{(\alpha)} - 2\theta \sum_{i=2}^{n+1} i^{\alpha}\xi_i - \theta^2 \sum_{i=2}^{n+1} i^{\alpha}.$$

The endpoint convention differs by one observation, and $(n+1)^{2\alpha} \sim n^{2\alpha}$ for fixed α.

After centering, the oracle score is a quadratic polynomial in the independent standard Gaussians. Its covariance with each ξ_i is zero because all centered Gaussian third moments vanish. The constant bias term has no variance. Consequently the exact variance transfer is

$$\mathrm{Var}(\Xi_{n,w}) = \mathrm{Var}(\Xi_{n+1}^{(\alpha)}) + 4\theta^2 \sum_{i=2}^{n+1} i^{2\alpha}.$$

This is the orthogonality argument of Analysis 09, Appendices D.1–D.2. Analysis 06's equation (4) and the proof of its proposition, equation (5), already establish $\mathrm{Var}(\Xi_{n+1}^{(\alpha)}) \sim 2n^{2\alpha}/(1+\alpha)$. We use that result rather than repeating its coefficient calculation.

When $n\theta^2 \to c \geq 0$, Appendix B.1 gives

$$\frac{\mathbb{E}\Xi_{n,w}}{n^{\alpha}} \to \frac{1}{\alpha} - \frac{c}{\alpha+1}, \qquad \frac{\mathrm{Var}(\Xi_{n,w})}{n^{2\alpha}} \to \frac{2}{\alpha+1} + \frac{4c}{2\alpha+1}.$$

For $c \neq \lambda$, the limiting mean coefficient is nonzero, so the first limit can be expressed with $\sim$ as in Section 4. At $c = \lambda$, only $\mathbb{E}\Xi_{n,w} = o(n^{\alpha})$ follows; we do not use an asymptotic equivalence to zero.

C.2 General lower bound and the boundary

The same Gaussian orthogonality works for any deterministic weights, so (1) holds whenever the mean is nonzero. Cauchy–Schwarz on the n weights gives

$$(\sum_{i=2}^{n+1} w_i)^2 \leq n \sum_{i=2}^{n+1} w_i^2, \qquad n\frac{\sum_{i=2}^{n+1} w_i^2}{(\sum_{i=2}^{n+1} w_i)^2} \geq 1.$$

For $c \neq \lambda$, $n(r_n - \theta^2) \to \lambda - c \neq 0$, so the mean is eventually nonzero. Rewriting (1) yields

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \geq \frac{4n\theta^2}{\{n(r_n - \theta^2)\}^2} \left[n\frac{\sum_{i=2}^{n+1} w_i^2}{(\sum_{i=2}^{n+1} w_i)^2} \right].$$

The prefactor tends to $4c/(\lambda - c)^2$, proving Proposition 4.1. For $c = \lambda$, its numerator tends to $4\lambda > 0$ and its denominator tends to zero. The bracket is at least one, so the ratio diverges along any sequence on which it is defined. At exact mean-zero points, division by the squared mean is undefined. This proves the boundary conclusion without requiring a rate for the smaller-order mean.

C.3 Full power-weight ratio

For $c \neq \lambda$, division of the limits in Appendix C.1 gives

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \to \frac{2/(\alpha+1) + 4c/(2\alpha+1)}{\{1/\alpha - c/(\alpha+1)\}^2}.$$

Since $1/\alpha = (\lambda)/(\alpha+1)$, multiplying numerator and denominator by $(\alpha+1)^2$ makes the denominator $(\lambda - c)^2$ and the numerator

$$2(\alpha+1) + \frac{4c(\alpha+1)^2}{2\alpha+1} = \frac{2\lambda}{\lambda-1} + \frac{4c\lambda^2}{\lambda^2-1}.$$

This proves (2), with a finite strictly positive limit. The boundary follows from Appendix C.2; equivalently, at $c = \lambda$ the variance has a positive $n^{2\alpha}$ coefficient whereas the squared mean is $o(n^{2\alpha})$. For $\alpha = 1$ and $\lambda = 2$, the mean is $\sim n(1 - c/2)$ away from $c = 2$, the variance is $\sim n^2(1 + 4c/3)$, and the ratio limit is $(4 + 16c/3)/(2 - c)^2$, agreeing with (2).

C.4 Comparison of asymptotic regimes

If $n\theta^2 \to 0$ with fixed $\alpha > 0$, the bias contribution to the mean and the linear variance are negligible on the scales n^α and $n^{2\alpha}$. Appendix C.1 therefore recovers the Analysis 06 ratio $2\alpha^2/(1+\alpha)$. For fixed $\lambda > 1$, the constant in (2) tends to this value as $c \downarrow 0$, but diverges as $c \to \lambda$ because its numerator remains positive while the denominator vanishes.

At fixed $c > 0$, $c \neq 1$, both numerator terms in (2) diverge as $\lambda \downarrow 1$, while $(\lambda - c)^2 \to (1-c)^2 > 0$. This recovers the qualitative obstruction at Analysis 09's alignment boundary. At $c = 1$, the denominator also vanishes and the constant again diverges. As $\lambda \to \infty$ with $c > 0$ fixed, the numerator tends to $2 + 4c$ and the denominator grows as λ^2, so the constant tends to zero. In this comparison $n/n_{\mathrm{RV},w} \to c/\lambda$ in the preceding fixed-parameter limit, placing the evaluation increasingly far before the weighted crossing.

All parameter comparisons follow the fixed-α, fixed-c asymptotics; no uniform or simultaneous limit is asserted. In particular, neither the small oracle benchmark nor a bounded square-weight factor gives a vanishing ratio for a fixed finite λ and fixed $c > 0$. Nor does a nonvanishing Chebyshev ratio by itself establish a positive limiting wrong-selection probability.